

Rishi Das

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Education

NIT Rourkela

B.Tech.+M.Tech.(Dual Degree), Mining [GPA: 8.35/10.0]

Rourkela, Odisha

Nov. 2022 – June 2027

Salt Lake School(E.M)

10+2 High School Education [GPA: 10/10]

Kolkata, West Bengal

April 2008 – June 2022

Relevant Coursework

- Mine Development
- Mine Surveying
- Mine Economics
- Mining Machinery
- UDG Metal Mining
- Mine Ventilation
- Rock Mechanics
- Computer Applications

Experience

Undergraduate Research Intern

May 2025 – Sept 2025

IIT Kharagpur

Kharagpur, West Bengal

- On-site internship entitled '**Embedded Systems Design and Coding**' under **Prof. Arabinda Routray at IIT KGP**. Worked on a research project titled "**Contactless Monitoring of Human Vitals and Emotions**" based on fluctuations in heart rate, respiratory rate, and other vital parameters, determined through non-invasive biomarkers using FMCW radar and thermal camera images. [Project Certificate](#)

Vocational Training

Jan 2025

Uranium Corporation of India Limited (UCIL)

Turamdih, Tatanagar

- Visited both Underground uranium mines and Open Cast mines and witnessed the **Shovel Dumper system, bench design, and drilling**.
- Observed the **Coal Handling Plant(CHP)**, including crushing, screening and loading system for dispatch.

Financial Portfolio Management Intern

Dec 2023 – Feb 2024

Finlatics

Remote

- Completed a **three-month virtual internship involving simulated trading via the Finlatics app**, supported by tutorial videos and mentor guidance; **achieved 80–90% accuracy** in market predictions.
- Earned multiple recognitions: [Performance Report](#), [Certificate of Completion](#), [Letter of Recommendation](#).

Projects

LST Image Extraction | *LANDSAT, SENTINEL, PyTorch, QGIS, Pix4DMapper, Python*

Aug 2025 - Present

- Spearheaded a remote sensing project to predict coal fires by applying a **super-resolution model (like DCN and ESRGAN)** to upscale the satellite image, calculating **Land Surface Temperature (LST)**, and validating thermal hotspots against drone data for enhanced mine safety and resource protection. Working on a ground dataset from **West Bokaro coal field in Ramgarh district, Jharkhand, India** and satellite data from different satellite missions like **LANDSAT, ASTER, MODIS and SENTINEL**.

Slope Stability Simulation | *Excel, RocScience Slide2, RS2, Ventsim, Ansys, Python, MATLAB*

Jul 2025 - Oct 2025

- Worked on MS Excel using formulas and charts for analysis on slope stability by different methods like the **Limit Equilibrium method, Finite Equilibrium Method**. Developed different material works on different systems of slopes, **calculated FOS for slope, and used bolting as a support to prevent failure**. Used MATLAB and Python for algorithms to calculate complex formulas faster, like **Neural Networks, Machine Learning and Fuzzy Inferences**. [MATLAB Certificate](#)

Technical Skills

Technical tools: Java, Python, HTML/CSS, MATLAB, Numpy, Pandas, MS Excel, MS PowerPoint

Mining Softwares: ANSYS, VENTSIM, SLIDE2, RS2, QGIS, USGS Earth Explorer

Leadership / Extracurricular

Secretary

May 2024 – April 2025

Finance Club, NITR

NIT Rourkela

- Organised 5 large events, including Finotsav, securing major sponsors and collaborations with companies like Unstoppable, Groww, StockEdge, and the CFA Institute.

Guest Lecture and Sponsorship Coordinator

October 2023 – December 2023

Innovision 2023

NIT Rourkela

- Led a team of 20+ people in Guest lectures and sponsorship. Gained experience in group work, sponsorships, and professional mailing.